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REVIEW ARTICLE

Potential Pharmacokinetic Herb-Drug Interactions: Have We Overlooked the Importance of Human Carboxylesterases 1 and 2?

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Abstract: Background: Herbal products have grown steadily across the globe and have increasingly been incorporated into western medicine for healthcare aims, thereby causing potential pharmacokinetic Herb-drug Interactions (HDI) through the inhibition or induction of drug-metabolizing enzymes and transporters. Human Carboxylesterases 1 (CES1) and 2 (CES2) metabolize endogenous and exogenous chemicals including many important therapeutic medications. The growing number of CES substrate drugs also underscores the importance of the enzymes. Herein, we summarized those potential inhibitors and inducers coming from herbal constituents toward CES1 and CES2. We also reviewed the reported HDI studies focusing on herbal products and therapeutic agents metabolized by CES1 or CES2.

Methods: We searched in PubMed for manuscript published in English after Jan 1, 2000 combining terms “carboxylesterase 1”, “carboxylesterase 2”, “inhibitor”, “inducer”, “herb-drug interaction”, “inhibitory”, and “herbal supplement”. We also searched specific websites including FDA and EMA. The data of screened papers were analyzed and summarized.

Results: The results showed that more than 50 natural inhibitors of CES1 or CES2, including phenolic chemicals, triterpenoids, and tanshinones were found from herbs, whereas only few inducers of CES1 and CES2 were reported. Systemic exposure to some commonly used drugs including oseltamivir, irinotecan, and clopidogrel were changed when they were co-administered with herb products such as goldenseal, black cohosh, ginger, St. John's Wort, curcumin, and Chinese compound formula in animals.

Conclusion: Nonclinical and clinical studies on HDIs are warranted in the future to provide safety information toward better clinical outcomes for the combination of herbal products and conventional drugs.

Keywords: Carboxylesterase (CES), CES1, CES2, herb-drug interactions, pharmacokinetics.

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1. INTRODUCTION

In Asia, herbal medicines have been used for many centuries. In western society, however, it's difficult to accurately estimate the prevalence of herbal products usage and co-administration with conventional medicines, because consumers of herbal products seldom report to their health care providers [1]. Only about 20% of the population acknowledges taking herbal products [2]. Obviously, this percentage is probably even higher in a population with medical conditions such as chronic gastrointestinal disorders, chronic pain, insomnia, liver disease, depression, asthma, and menopause [1, 3]. Indeed, the herbal supplement sales in America increased by 7.7% in 2016, which marks the 13th consecutive year of overall sales growth of herbal supplements and the first time that the total US retail sales of these products have surpassed to \$7 billion [4]. Meanwhile, the global market of herbal supplements is expected to reach \$107 billion by 2017 (http://www.prweb.com/releases/herbal_supplements/herbal_remedies/prweb9260421.htm). Increased sales of herb products and underestimated self-administered practice of herbal supplements raise concerns about potential probabilities of unexpected herb-drug interactions (HDIs).

HDIs can be defined as a pharmacological or clinical response to the administration or co-exposure of a conventional medication and an herbal medicine beyond that which is expected from the

known effects of each substance given individually [5, 6]. The underlying mechanisms could be some alternations of the co-administered drug's ("victim") pharmacodynamics and/or pharmacokinetics caused by an herbal product ("perpetrator"). The resulting alterations potentially lead to some efficacy or drug-related toxicity problems. From the point of view of pharmacokinetics, inhibiting and/or inducing the drug-metabolizing enzymes and drug transporters by herbal supplements would lead to HDI. It is relevant to conduct HDI studies aiming to identify and prevent adverse drug interactions prospectively, as well as to modulate potentially beneficial interactions. Through HDI studies, we will be able to perform much more efficient management of HDIs in the future.

As members of the α/β -hydrolase-fold family, human carboxylesterases (CESs) efficiently hydrolyze various xenobiotics and endobiotics. In humans, CES1, CES2, CES3, CES5, and CES6 have been reported [7], but the majority of CESs has been segregated into the CES1 and CES2 families [7-11]. The expression of human carboxylesterases is generally ubiquitous, but human liver predominantly contains CES1 with smaller quantities of CES2, whereas the small intestine contains CES2 with virtually no CES1 [12, 13]. Except for endogenous acyl-glycerols, long-chain acyl-carnitine, and long-chain acyl-coenzyme A, CES1 and CES2 also involve in the activation or inactivation of some important prodrugs and drugs (Table 1), such as antiplatelet agents (e.g., clopidogrel, aspirin, and dabigatran), angiotensin-converting enzyme inhibitors (ACEI, e.g., enalapril, trandolapril, and imidapril), antiviral agents (e.g., oseltamivir), immunosuppressive agents (e.g., mycophenolate mofetil), CNS agents (e.g., methylphenidate) and anticancer drugs (e.g., irinotecan and capecitabine). The general

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Table 1. Substrate drugs of CES1 and CES2*.

CES1 Substrate Drugs	CES2 Substrate Drugs
Antiplateletes/Anticoagulants	Antiplateletes/Anticoagulants
Clopidogrel	Acetylsalicylic acid
2-oxo-clopidogrel	Prasugrel
Dabigatran etexilate	Dabigatran etexilate
Angiotensin-converting enzyme inhibitors	Angiotensin receptor blockers
Enalapril, Imidapril	Candesartan cilexetil
Benazepril, Ramipril	Olmesartan medoxomil
Trandolapril, Quinapril	Azilsartan medoxomil
Temocapril, Cilazapril	Antiviral agents
Delapril, Perindopril, Moxeipril, Fosinopril	Tenfovir disoproxil
Antihypertensive agents	Adefovir dipivoxil
Sacubitril	Valacyclovir
Antihyperlipidemic agents	CNS agents
Simvastatin	Cocaine
Lovastatin	Heroin
Clofibrate	6-monoacetylmorphine
Fenofibrate	Immunosuppressive agents
Antiviral agents	Methylprednisolone sodium succinate
Oseltamivir	Deflazacort
Sofosbuvir	Oncology agents
Tenofovir alafenamide	Irinotecan
CNS agents	Capecitabine
Methylphenidate	Anesthetic drug
Cocaine	Procaine
Meperidine	
Flumazenil	
Rufinamide	
Immunosuppressive agents	
Mycophenolate mofetil	
Ciclesonide	
Oncology Agents	
Nintedanib	
Capecitabine	
Tryptophan hydroxylase inhibitor	
Telotristat ethyl	
Analogues of prostaglandin	
Travoprost	
Antispasmodic	
Oxybutynin	
Others	
Dimethyl fumarate	

* Data from FDA, EMA and published articles [13, 40].

metabolic scheme is shown in Fig. (1). Generally speaking, CES1 enzyme prefers esters with a large, bulky acyl group and a small alcohol group (Supplemental Fig. 1), whereas CES2 has the opposite preference, substrates with a small acyl group and a large alcohol group (Supplemental Fig. 1) [12, 13].

The purpose of this mini-review is to 1) provide a brief overview of inhibitors or inducers towards CES1 and CES2 from herbal supplements and herbal medicines; and 2) assess the potential HDIs between herb and therapeutic agents metabolized by CES1 and/or CES2.

2. THE GENERAL CONSTITUENTS OF HERBAL SUPPLEMENTS AND HERBAL MEDICINES

Unlike the western drugs with clear chemicals and structures, the constituents of herbal supplements and herbal medicines are very complex, partly due to the species of plants used in the supplements, the sources of the herb, preparation procedure, formulation, and even the standardization process [14]. According to the market report issued by American Botanical Council in 2017, the major constituents of the 40 top-selling herbal supplements in 2016 are flavonoids, triterpenoids, alkaloids, and fatty acids (Supplemental Table 1).

Volume I of the Chinese Pharmacopoeia (Pharmacopoeia Commission of the People's Republic of China, 2010) covers 474 monographs of commonly used herbs. Among them, the qualities of the 284 individual herb are encouraged to be controlled using High-Performance Liquid Chromatography (HPLC) techniques. Interestingly, more than 60% marker compounds monitored by HPLC for quality control are polyphenols and saponins [15].

3. INHIBITORY ROLES OF HERBAL CONSTITUENTS TOWARDS CES1 AND CES2

3.1. Phenolic Chemicals

Flavonoids are a group of diverse, low-molecular-weight polyphenolic compounds occurring in the herbal supplements and herbal medicines. These bioactive constituents possess unique biological properties that may be responsible for various health benefits. *Fructus psoraleae* is the dried ripe seeds of *Psoralea corylifolia* L. (Fabaceae) and widely used in Asia for the treatment of various diseases including asthma, diarrhea, and osteoporosis. Recently, scientists found that some flavonoids occurring in the *Fructus psoraleae* have inhibitory roles against both CES1 and CES2. Among them, neobavaisoflavone, corylifolinin, coryfolin and corylin non-competitively inhibited the activity of CES1, whereas bavachinin competitively inhibited the activity of CES1. The inhibition of kinetic parameter (K_i) was calculated to be 5.3, 9.4, 1.9, 0.7 and 0.5 μM for neobavaisoflavone, corylifolinin, coryfolin, corylin and bavachinin (Table 2), respectively [16]. Meanwhile, neobavaisoflavone, isobavachalcone, bavachinin, corylifol A and bakuchiol were found to be potent inhibitors of CES2, with low K_i values ranging from 0.62 to 3.89 μM (Table 2) [17]. More recently, Liu and colleagues identified three bioactive flavonoids including sanggenone D, kuwanon G and sanggenone C from White Mulberry Root-bark, which is an edible Chinese herb used for the treatment of inflammation, nephritis and asthma. These three flavonoids were found to non-competitively inhibit the CES2-mediated hydrolysis with K_i values ranging from 0.76 to 1.09 μM [18].

Wang and colleagues identified 10 phenolic acid glycosides from *E. ebracteolata*, including γ -pyrone-2-O- β -d-(6'-galloyl)-glucopyranoside, γ -pyrone-2-O- β -d-(6'-digalloyl)-glucopyranoside, pyromeconic acid-3-O- β -d-glucopyranoside-6'-(O-4''-hydroxybenzoate), scopoletin-7-O- β -d-(6'-galloyl)-glucopyranoside, benzoic acid-4-O- β -d-(6'-galloyl)-glucopyranoside, gallic acid 4-O- β -d-(6'-O-galloyl)-glucopyranoside and, gallic acid 3-O- β -d-(6'-O-galloyl)-glucopyranoside, corilagin, ellagic acid-4-O- β -d-glucopyranoside, and brevifolin [19]. These herbal constituents displayed inhibitory roles to some extent against CES2 with mean IC_{50} values ranging

General metabolic scheme of drugs catalyzed by CES1 or CES2

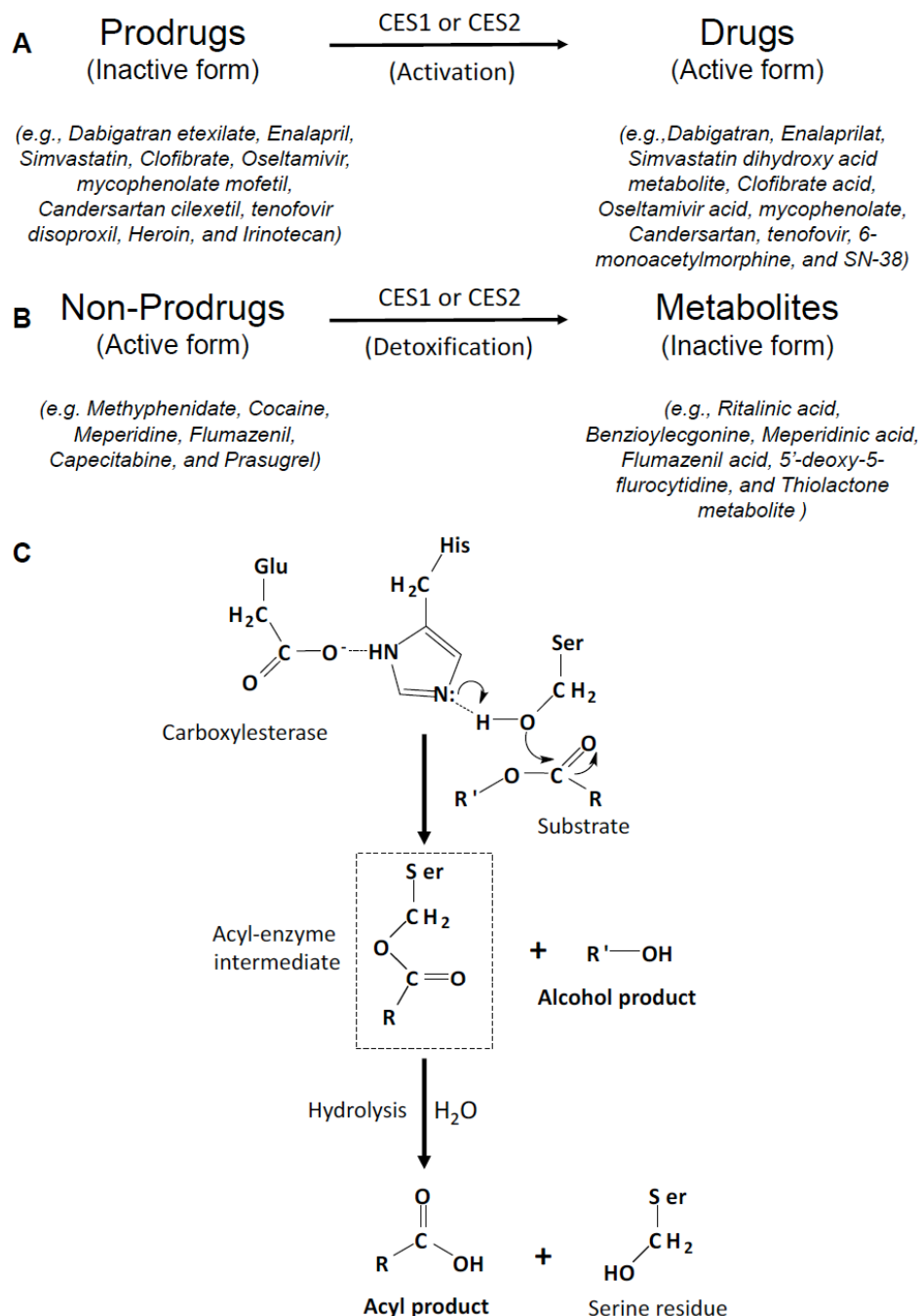


Fig. (1). General metabolic scheme of drugs catalyzed by CES1 and/or CES2. **(A).** CES1 or CES2 catalyzes prodrugs into active drugs; **(B).** CES1 or CES2 inactivates non-prodrugs forming different metabolites. **(C).** The catalytic mechanism of human carboxylesterase. The enzyme active site serine hydroxyl group attacks the carboxyl-ester linkage in the substrate to generate the alcohol product and the covalent acyl-enzyme intermediate, which is then attacked by water to form acyl product and release serine residue to its original state.

from 7.17 μM to more than 100 μM (Table 2). Notably, the scopoletin-7-O- β -d-(6'-galloyl)-glucopyranoside, a coumarin glucoside, obviously inhibited CES2 with an IC_{50} value 7.17 μM , suggesting that the galloyl groups may enhance the inhibitory effects on CES2 for these isolated compounds.

3.2. Triterpenoids

Triterpenoids are derivatives of triterpene molecules, including oleanolic acid, ursolic acid, moronic acid, and betulinic acid, which

are widely distributed in herbal products and herbal medicines. *Alisma orientale* (Sam.) Juzep. (*Alismataceae*) is a traditional Chinese medicinal herb. Terpenoids including protostane triterpenoids and guaiane sesquiterpenoids have been identified as *A. orientale*'s characteristic constituents. Recently, Mai and colleagues observed that protostane triterpenoids alismanol B, 25-O-ethylalisol A, alismanol D, alismanol F, and 11-deoxyalisol A displayed inhibitory roles toward CES2, with IC_{50} values of 8.68, 4.72, 4.58, 2.02, and 4.63 μM (Table 2), respectively. The K_i value of alismanol

Table 2. Potential natural inhibitors towards CES1 and CES2.

Natural Inhibitors	Inhibition Potency				References
	Towards CES1		Towards CES2		
	<i>IC</i> ₅₀ (μM)	<i>K</i> _i (μM)	<i>IC</i> ₅₀ (μM)	<i>K</i> _i (μM)	
<i>Phenolic chemicals</i>					
Bavachinin	-	0.5	4.31	1.12	[16,17]
Coryfolin	-	1.9	-	-	[16]
Corylin	-	0.7	-	-	[16]
Coryfolinin	-	9.4	-	-	[16]
Neobavaisoflavone	-	5.3	6.39	3.89	[16,17]
Isobavachalcone	-	-	2.85	1.61	[17]
Corylifol A	-	-	0.87	0.62	[17]
Bakuchiol	-	-	7.28	2.12	[17]
Kuwanon G	-	-	1.04	1.09	[18]
Sanggenone C	-	-	1.02	0.76	[18]
Sanggenone D	-	-	1.09	0.97	[18]
γ-Pyrone-2-O-β-D-(6′-galloyl)-glucopyranoside	-	-	43.3	-	[19]
γ-Pyrone-2-O-β-D-(2′,6′-digalloyl)-glucopyranoside	-	-	41.4	-	[19]
Pryomeconic acid-3-O-β-D-glucopyranoside-6′-(O-4″-hydroxybenzoate)	-	-	17.5	-	[19]
Scopoletin-7-O-β-D-(6′-galloyl)-glucopyranoside	-	-	7.17	-	[19]
Benzoic acid-4-O-β-D-(6′-galloyl)-glucopyranoside	-	-	82.4	-	[19]
Gallic acid 4-O-β-D-(6′-O-galloyl)-glucopyranoside	-	-	≥100	-	[19]
Gallic acid 3-O-β-D-(6′-O-galloyl)-glucopyranoside	-	-	≥100	-	[19]
Corilagina	-	-	99.5	-	[19]
Ellagic acid-4-O-β-D-glucopyranoside	-	-	≥100	-	[19]
Brevifolin	-	-	≥100	-	[19]
Paeoveitol B	-	-	≥100	-	[19]
<i>Triterpenoids</i>					
Alismanol B	-	-	8.68	-	[20]
25-O-ethylalisol A	-	-	4.72	-	[20]
Alismanol D	-	-	4.58	-	[20]
Alismanol F	-	-	2.02	1.76	[20]
11-Deoxyalisol A	-	-	4.63	-	[20]
Glycyrrhetic acid	21.0	-	69.3	-	[21]
11-Deoxo-glycyrrhetic acid	10.5	-	6.95	-	[21]
Glycyrrhetic acid	26.3	-	5.70	-	[21]

Table (2) contd....

Natural Inhibitors	Inhibition Potency				References
	Towards CES1		Towards CES2		
	IC ₅₀ (μM)	K _i (μM)	IC ₅₀ (μM)	K _i (μM)	
3-O-(β-Carboxypropionyl)-11-deoxy-glycyrrhetic acid-30-ethyl ester	20.4	-	0.02	-	[21]
Oleanolic acid	0.28	-		-	[22]
Ursolic acid	0.24	-		-	[22]
Compound 12	-	-	28.1	-	[19]
Compound 13	-	-	25.3	-	[19]
Euphorbic acid	-	-	70.2	-	[19]
Euphorbin A	-	-	≥100	-	[19]
Euphorbin B	-	-	≥100	-	[19]
Euphorbin C	-	-	≥100	-	[19]
Tanshinones					
Dihydrotanshinone	-	0.39	-	0.12	[24]
Miltirone	-	0.39	-	0.12	[24]
Cryptotanshinone	-	0.54	-	0.54	[24]
Tanshinone IIA	-	6.89	-	2.45	[24]
Tanshinone I	-	26.3	-	14.6	[24]
Tanshinone IIA sulfonate	-	100	-	3.90	[24]
Fatty acids from herb					
Arachidonic acid	2	1.7	-	-	[26]
Linoleic acid	9	-	-	>100	[26]
Γ-linoleic acid	19	-	-	>100	[26]
Myristic acid	9	-	-	>100	[26]
Myristoleic acid	12	-	-	>100	[26]
Palmitic acid	25	-	-	>100	[26]
Palmitoleic acid	7	-	-	>100	[26]
Oleic acid	7	-	-	>100	[26]
11,12-Epoxy-5E,8E,14E- eicosatrienoic acid	27	-	-	-	[26]
14,15-Epoxy-5E,8E,11E- eicosatrienoic acid	38	-	-	-	[26]
15-Deoxy-Δ ^{12,14} -prostaglandin J ₂	12	-	-	-	[26]

F toward CES2 was observed to be 1.76 μ M using a mixed inhibition model [20].

Glycyrrhetic acid is a triterpenoid aglycone component of the natural product glycyrrhizic acid, which is abundant in licorice root, an important herb in traditional Chinese medicine. Glycyrrhetic acid is a weak inhibitor toward CES1 and CES2 with IC_{50} values of 20.97 and 69.26 μ M, respectively [21]. However, some synthetic derivatives of glycyrrhetic acid are moderate to strong selective inhibitors against CES2. For example, the IC_{50} values against CES2 were 6.95, 5.70, and 0.02 μ M for 11-deoxy-

glycyrrhetic acid, glycyrrhetic acid, and 3-O-(β -carboxypropionyl)-11-deoxy-glycyrrhetic acid-30-ethyl ester, respectively. But the IC_{50} values against CES1 were 10.53, 26.26, and 20.41 μ M [21] (Table 2), respectively.

Zou and colleagues found that natural pentacyclic triterpenoids oleanolic acid and ursolic acid displayed potent inhibitory effects against CES1 (IC_{50} 0.28 and 0.24 μ M, respectively) (Table 2) with relatively higher selectivity over CES2 by 19-folds [22].

Euphorbia ebracteolata Hayata is a perennial herbaceous plant widely distributed in China. Its root is poisonous and is used in

traditional Chinese medicine for the treatment of many diseases such as cancer, swelling, and warts [23]. Very recently, Wang and colleagues identified 7 monoterpenoids from *E. ebracteolata* Hayata. These chemicals showed weakly inhibitory roles towards CES2 with IC_{50} values changing from 25.27 μ M to > 100 μ M [19] (Table 2).

3.3. Tanshinones

Tanshinones are ones of those major bioactive constituents from *Salvia miltiorrhiza* roots (Danshen), which is a very important herbal medicine and used extensively in the treatment of cardiovascular diseases especially for angina pectoris in China. Hatfield and colleagues found that tanshinones in Danshen could potently *in vitro* inhibit human CES1 and CES2, especially towards CES2 [24]. The K_i values were 0.39, 0.39, 0.54, 6.89, 26.25, and 100 μ M for dihydrotanshinone, miltirone, cryptotanshinone, tanshinone IIA, tanshinone I, and Tanshinone IIA sulfonate against CES1 (Table 2), respectively. However, these data changed to 0.12, 0.12, 0.54, 2.45, 14.55, and 3.90 μ M (Table 2), respectively, when CES2 was evaluated. Inconsistently, a Danshen extract containing tanshinones didn't alter the pharmacokinetics of clopidogrel after oral administration to rats (400 mg/kg extract) [25].

3.4. Other Herbal Constituents

Fatty acids are important dietary sources of fuel including plant oil and animal fat for humans because, when metabolized, they yield large quantities of ATP, which plays critical roles in energy metabolism and human health. The list of the 40 top-selling herbal supplements in 2016 in America (Supplemental Table 1) includes flax seed/flax oil, coconut oil, and chia seed/chia oil with a fatty acid such as linoleic acid as active and marker compounds. Crow and colleagues found that some natural fatty acids could strongly inhibit the hydrolytic activity of THP 1 cell lysate and recombinant CES1 [26]. The IC_{50} value was more than 100 μ M against CES2 for the tested fatty acids, including myristic acid, myristoleic acid, palmitic acid, palmitoleic acid, oleic acid, linoleic acid, γ -linoleic acid, and arachidonic acid (Table 2). However, these natural products are potent inhibitors against CES1, evidenced by the IC_{50} ranging from 2 to 38 μ M. Of particular, the most potent inhibitor, arachidonic acid, non-competitively suppressed CES1 with the K_i value of 1.7 μ M.

4. INDUCTION OF CES1 BY HERBAL CONSTITUENTS

CES1 could be induced by glucose [27]. Antioxidants such as sulforaphane compounds were also found to induce CES1 [28]. Sulforaphane (4-methylsulfinylbutyl isothiocyanate; 1-isothiocyanato-4-methylsulfinylbutane) is a dietary and herbal phytochemical that is present in plants as a biologically inactive precursor [29]. Chen and colleagues found that sulforaphane is a potent inducer of CES1 in various types of cells, indicating that the induction of CES1 by antioxidants represents a general phenomenon [28].

5. POTENTIAL HDIS BETWEEN HERBS AND AGENTS METABOLIZED BY CES1 AND/OR CES2

As a top-selling herbal supplement in the past years, Goldenseal [*Hydrastis canadensis* L. (*Ranunculaceae*)] has been used for centuries for the treatment of a variety of ailments such as urinary tract infections and internal bleeding. These biological activities are associated with the isoquinoline alkaloids hydrastine and berberine. Liu and colleagues found that the ethanol extract of goldenseal could inhibit the oseltamivir carboxylate formation by 75% in the pooled mixed gender human liver microsomes, displaying a strong inhibitory role against CES1 [30].

Gorman and colleagues found that black cohosh extract was the strongest inhibitor (K_i =1.62 mg/mL, competitive inhibition) which was tested on carboxylesterase-mediated biotransformation of iri-

notecan to its active metabolite SN-38 in human liver microsomes. Ginger root extract was a stronger inhibitor (IC_{50} = 25.5 mg/mL, indeterminate inhibition) compared with St John's wort (IC_{50} = 49.4 mg/mL, non-competitive inhibition) for the SN-38 formation catalyzed by carboxylesterase [31]. St John's wort was confirmed to enhance the pharmacodynamic response of clopidogrel [32, 33], and the inhibition of carboxylesterase by St. John's wort might play a critical role [31], because CES1 hydrolyzes the inactivation reaction of clopidogrel.

Curcumin, the major constituent of turmeric, has a variety of potentially beneficial health effects. Seven-day oral administration of curcumin 100 mg/kg to rats significantly increased the AUC and C_{max} of clopidogrel carboxylic acid by 1.61 and 1.81 times, respectively [34]. As a prodrug, clopidogrel has competitive metabolic pathways *in vivo* towards forming active metabolite H4 catalyzed by hepatic cytochrome P450 enzymes and inactive metabolite clopidogrel carboxylate hydrolyzed by CES1, which takes place in the liver and accounts for 85% of an administered dose [13]. So, the increased exposure (AUC and C_{max}) to clopidogrel acid might be partly due to the induction of the CES1 by curcumin.

Flos Lonicerae (JYH), *Folium Perillae* (ZSY), and the over-the-counter preparation such as *Radix isatidis* (BLG) are commonly taken with oseltamivir by the public combating influenza symptoms. Yang and colleagues observed that BLG exerted only a slight inhibition of oseltamivir carboxylate formation catalyzed by CES1, whereas JYH and ZSY could significantly suppress the activity of this enzyme [35].

Shengjiang Xiexin Decoction (SXD) is a classic Chinese medicine formula composed of 8 Chinese herbs, which is used to treat gastroenteritis, ulcerative colitis, and diarrhea. Rat studies conducted by Guan *et al.* showed that SXD pretreatment increased the AUC of irinotecan (CPT-11), a substrate drug of CES2. Consistently, the AUC and C_{max} of its active metabolite SN-38 produced by CES2 decreased. An *In vitro* study revealed that SXD significantly inhibited the CES2 expression in the SXD-pretreated rats [36].

6. INHIBITORY ROLES IN VITRO AND POOR ORAL BIOAVAILABILITY

Herbal constituents exert inhibitory roles towards CES1 and CES2 *in vitro* incubated with liver microsomes or recombinant human carboxylesterase, just as above summarized (Table 2). An important concern is the pre-systemic metabolism and oral bioavailability of these herbal constituents. Generally speaking, herbal constituents such as phenolic chemicals and triterpenoids display very poor oral bioavailability [37, 38]. It is not easy for these constituents after oral administration to achieve concentration levels that have been investigated under *in vitro* incubation conditions. As an inhibitor, its systemic concentration in the bloodstream ([I]) and inhibition constant (K_i) value obtained from the inhibitory kinetic study can be used to predict the potential risk of HDIs. The ranges of [I]/ K_i ratios for an inhibitor that would give rise to a low, medium, and high risk of pharmacokinetically significant interactions with substrates of CES1 or CES2 would be < 0.1, 0.1-1.0, and > 1.0, respectively [39]. Therefore, there is a minor possibility that phenolic chemicals and triterpenoids can cause serious carboxylesterase-mediated HDIs. However, some tanshinones including dihydrotanshinone, miltirone, and cryptotanshinone significantly inhibit the carboxylesterase-catalyzed transformation *in vitro* with K_i values ranging from 0.12 to 0.54 μ M. Moreover, herbal constituents might have higher pre-systemic concentrations in the stomach and small intestine after oral administration. Thus, the risk of Danshen-drug interactions probably increases especially for CES2-hydrolyzed drugs because CES2 is predominantly found in the small intestine.

CONCLUSION

Unlike CYP450 enzymes, data of human CES1 and CES2 inhibition and induction by potential herbal products including its constituents *in vitro* and *in vivo* are very limited. Based on the published data, the risk of pharmacokinetic HDI between the herbal supplement and therapeutic agents metabolized by human CES1 and CES2 is likely to be low. But the occurrence and significance of any specific HDIs depend on various factors, such as dose, frequency and timing of herb/drug intake, route of administration, and therapeutic range. Human CES1 and CES2 are versatile enzymes involving in the activation or inactivation of many very important medications. Thus, we need to conduct preclinical and clinical studies of potential HDIs that will provide useful information for both consumers, health care providers and herb suppliers aiming to achieve better clinical outcomes.

LIST OF ABBREVIATIONS

ACEI	=	Angiotensin-converting Enzyme Inhibitors
ATP	=	Adenosine Triphosphate
AUC	=	Area Under a Curve
BLG	=	Ban Lan Gen
CES1	=	Carboxylesterase 1
CES2	=	Carboxylesterase 2
C _{max}	=	Maximum Concentration
CNS	=	Central Nervous System
CPT-11	=	Irinotecan
CYP450	=	Cytochrome P450
HDI	=	Herb-drug Interaction
IC ₅₀	=	Half maximal Inhibitory Concentration
JYH	=	Jin Yin Hua
K _i	=	Inhibitory Constant
SXD	=	Shengjiang Xiexin Decoction
ZSY	=	Zi Su Ye

CONSENT FOR PUBLICATION

Not applicable.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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SUPPLEMENTARY MATERIAL

Supplementary material is available on the publisher's website along with the published article.

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